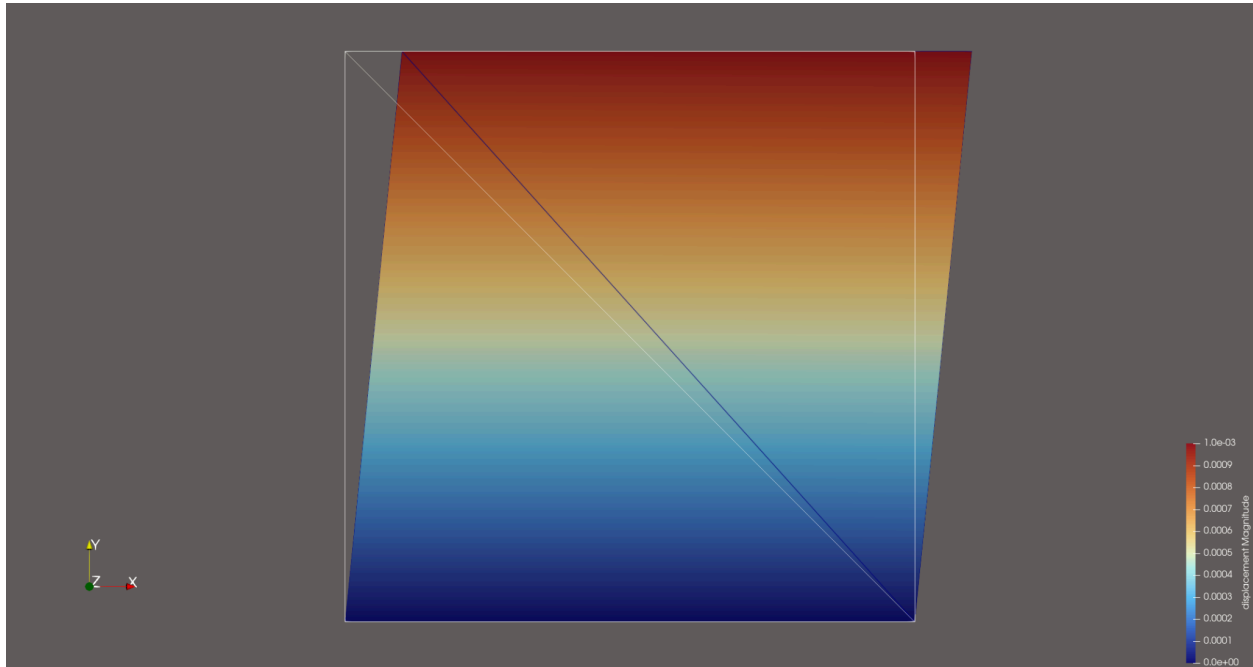
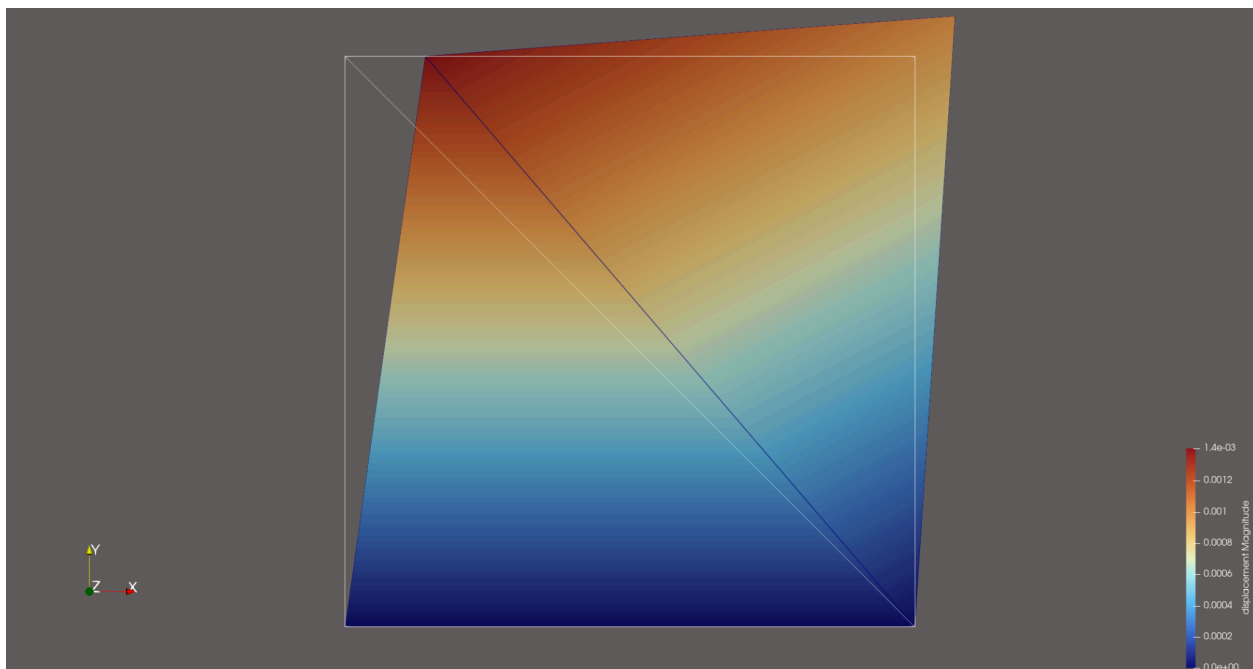


Part 1 (a): 1x1 Linear Triangle Elements with 1 Quad Point

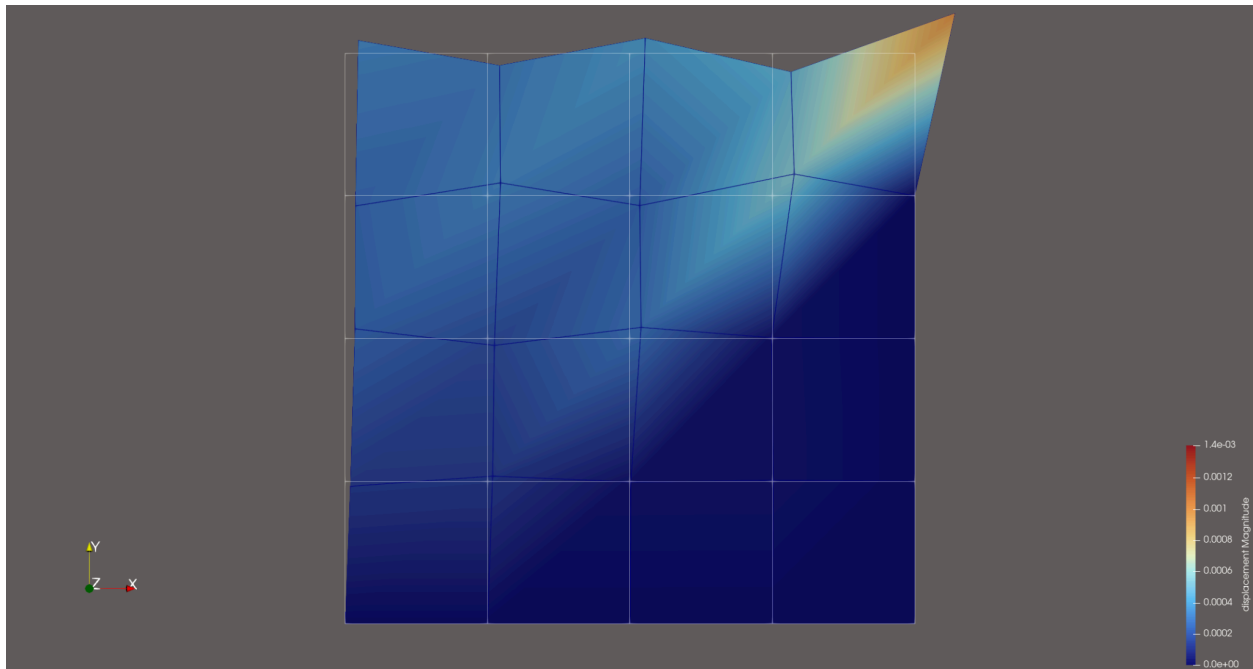


For BCs given in Fig1(a)



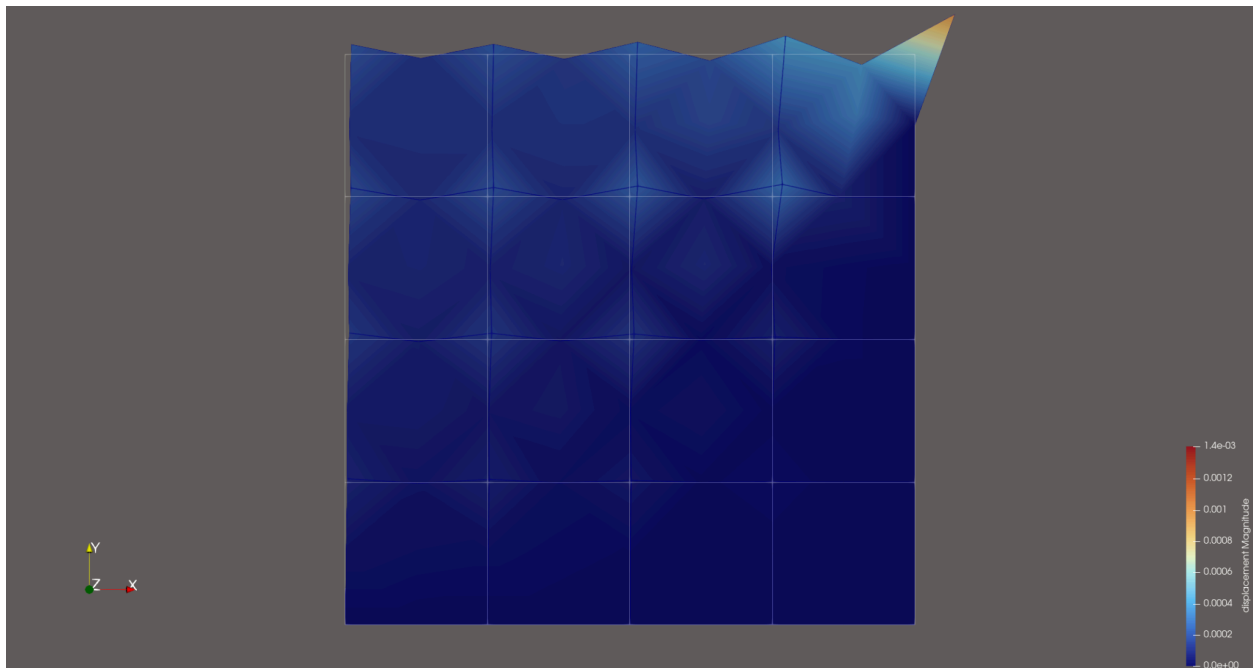
For BCs given in Fig1(b)

Part 1 (b): 4x4 Linear Quadrilateral Elements with 1 Quad Point



For BCs given in Fig1(c)

Part 1 (c): 4x4 Quadratic Quadrilateral Elements with 2 Quad Points

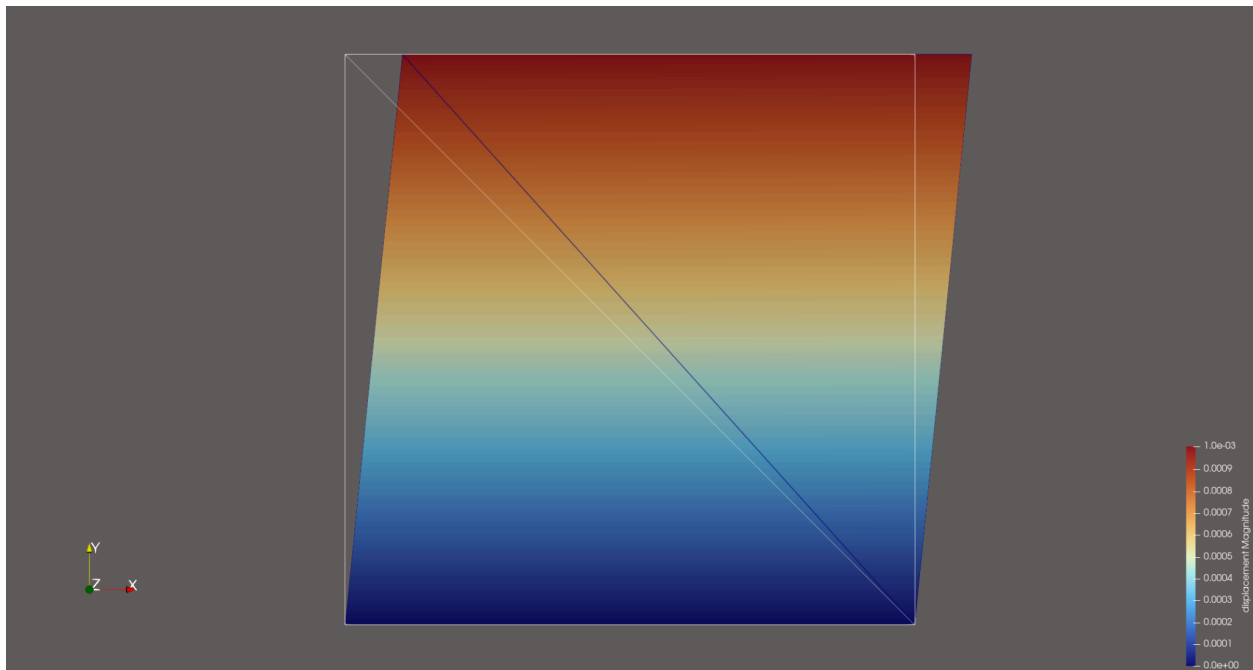


For BCs given in Fig1(c)

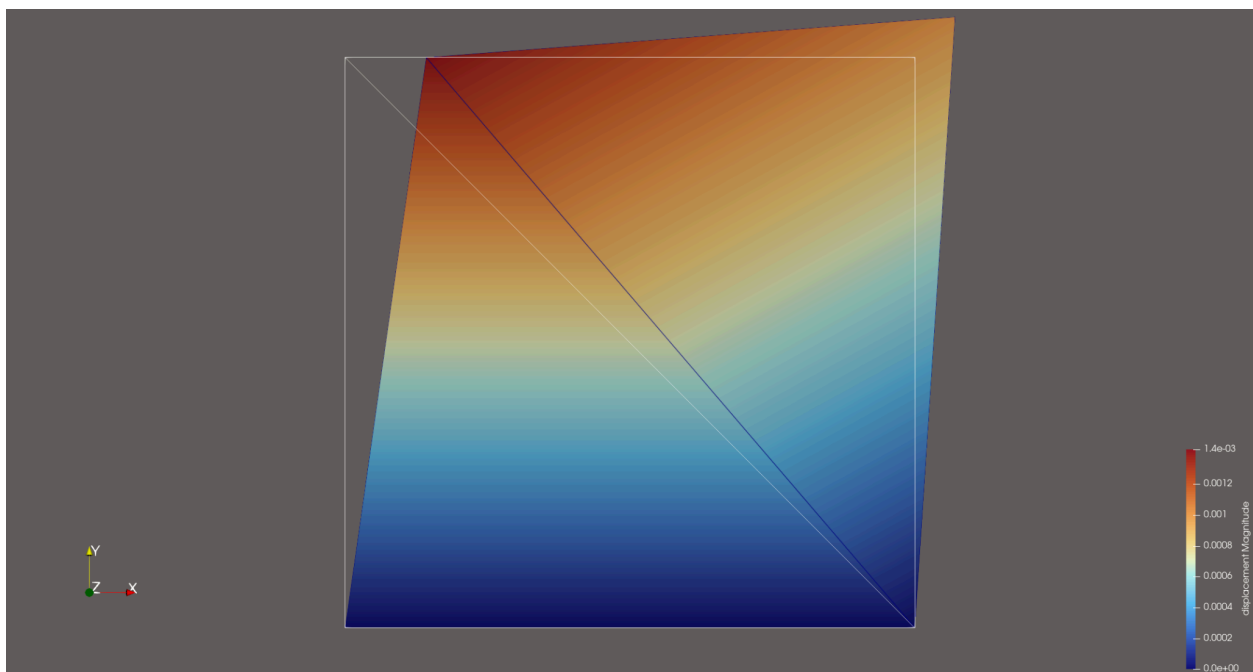
Part 2 (a): 1x1 Linear Triangle Elements

#u dofs = 3 ; quadRule = 1

#p dofs = 1 ; quadRule = 1



For BCs given in Fig1(a)

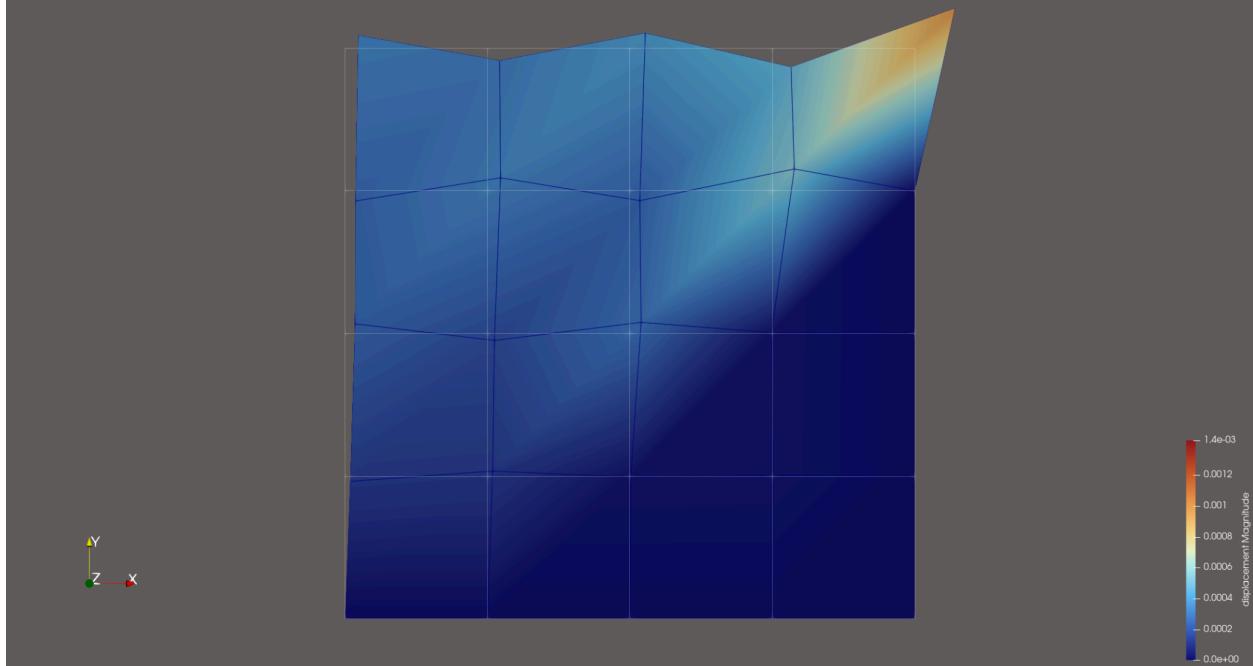


For BCs given in Fig1(b)

Part 2 (b): 4x4 Linear Quadrilateral Elements

#u dofs = 4 ; quadRule = 1

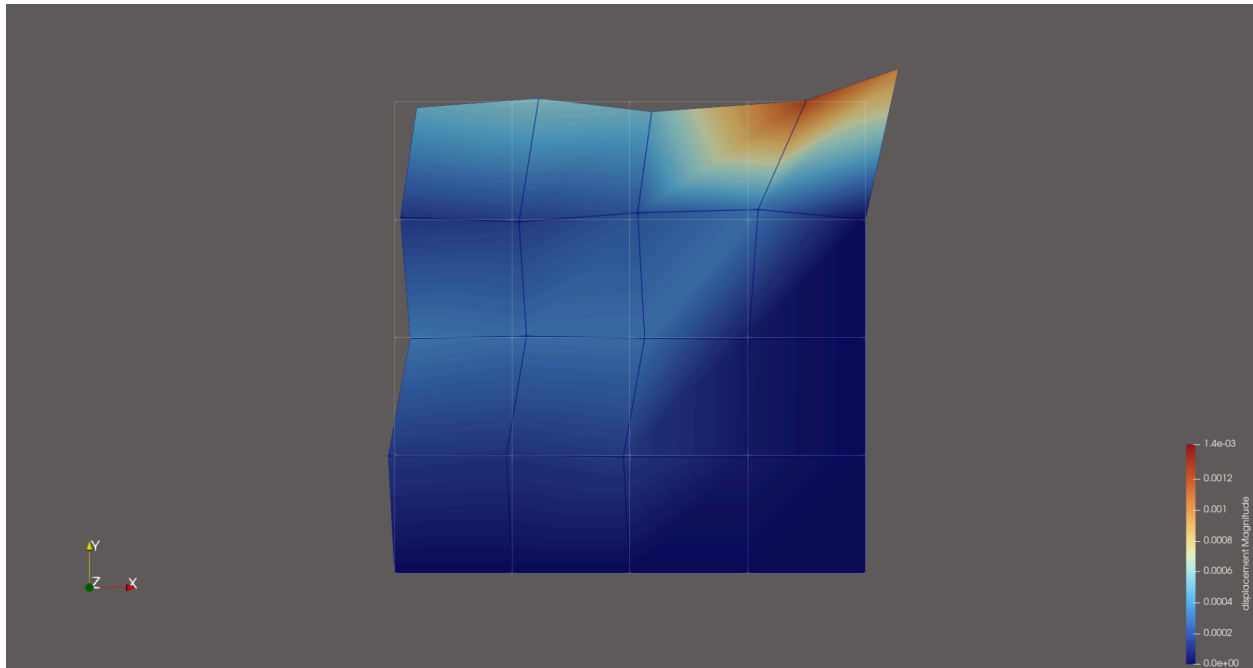
#p dofs = 1 ; quadRule = 1



For BCs given in Fig1(c)

#u dofs = 4 ; quadRule = 2

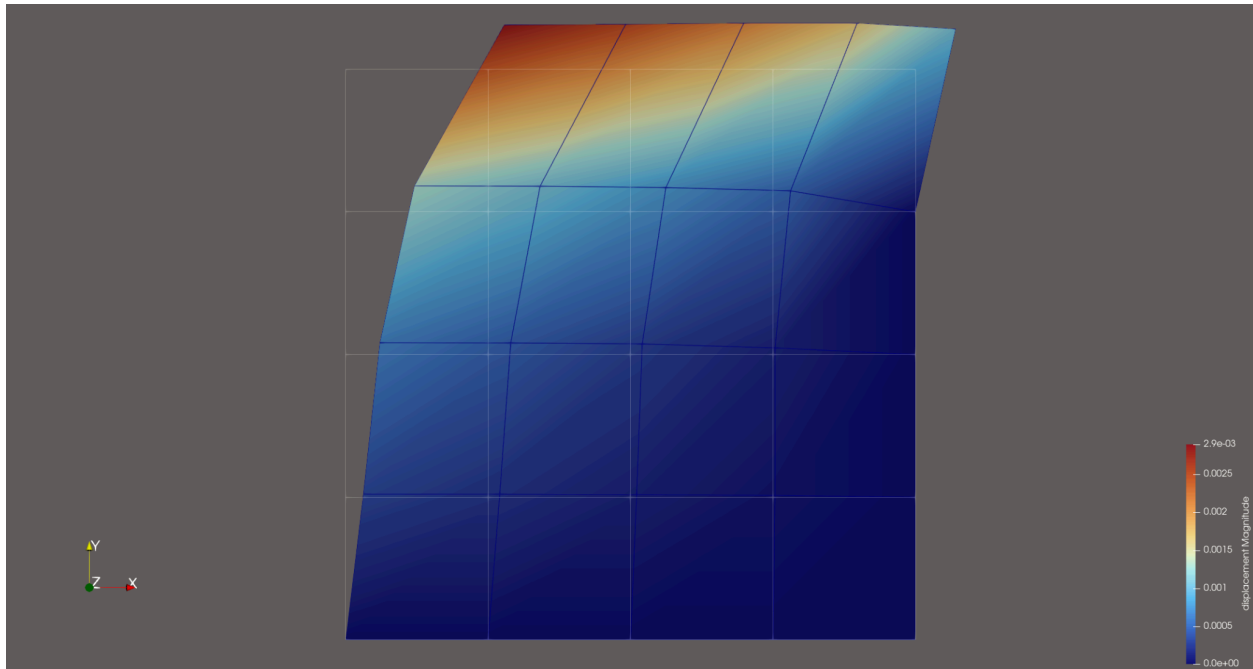
#p dofs = 4 ; quadRule = 2



For BCs given in Fig1(c)

#u dofs = 4 ; quadRule = 2

#p dofs = 4 (bubble) ; quadRule = 2

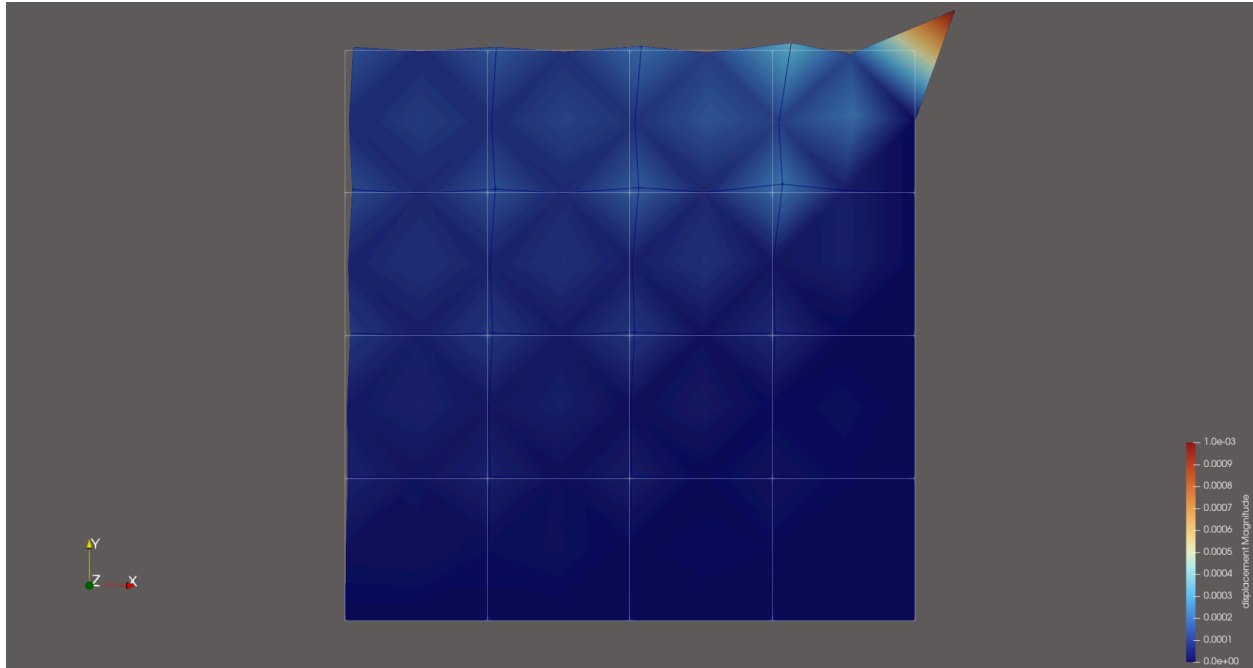


For BCs given in Fig1(c)

Part 2 (c): 4x4 Quadratic Quadrilateral Elements

#u dofs = 9 ; quadRule = 2

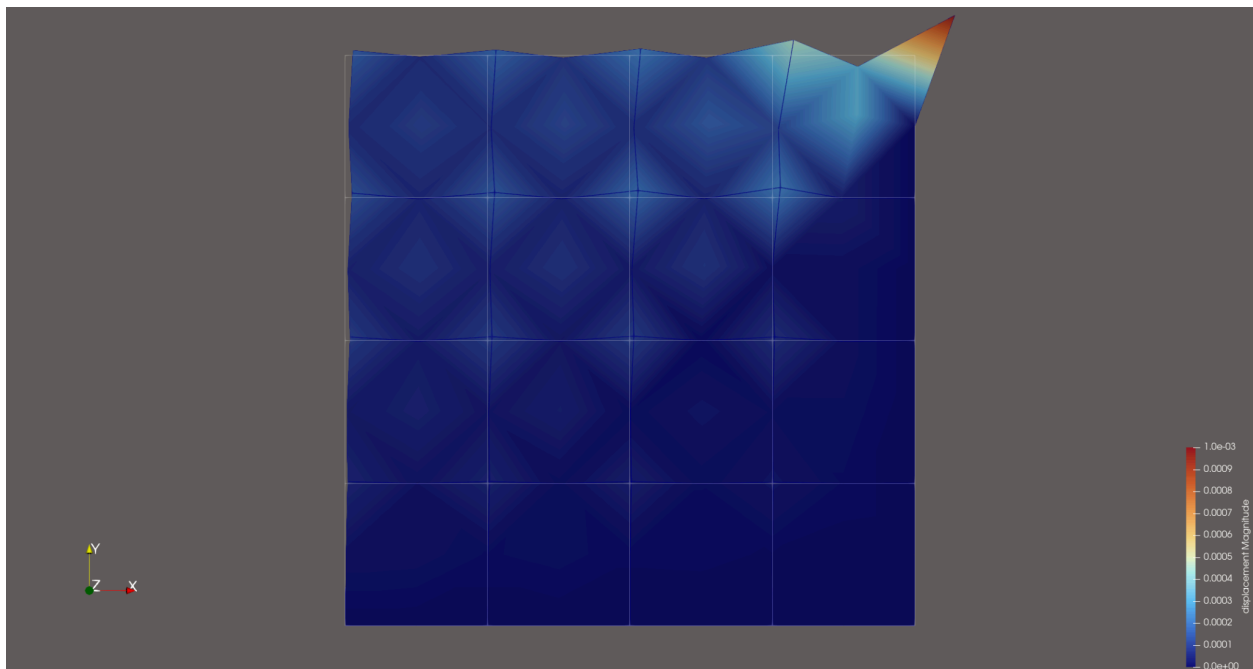
#p dofs = 1 ; quadRule = 1



For BCs given in Fig1(c)

#u dofs = 9 ; quadRule = 2

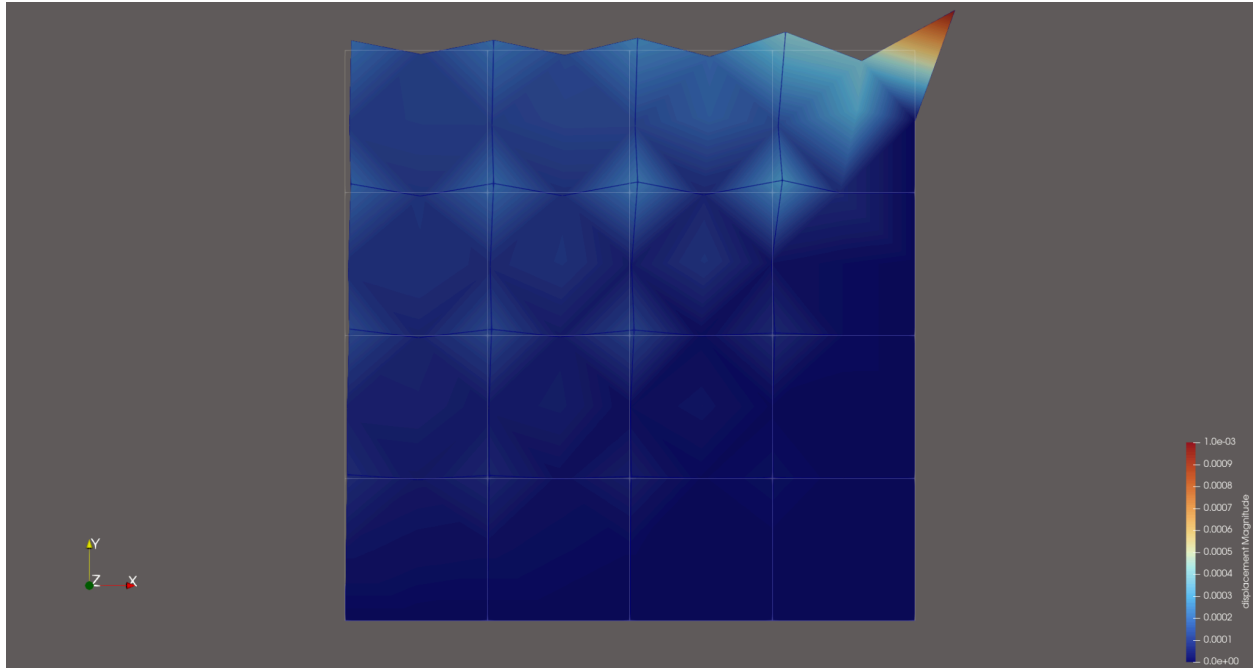
#p dofs = 4 ; quadRule = 2



For BCs given in Fig1(c)

#u dofs = 9 ; quadRule = 2

#p dofs = 4 (bubble) ; quadRule = 2

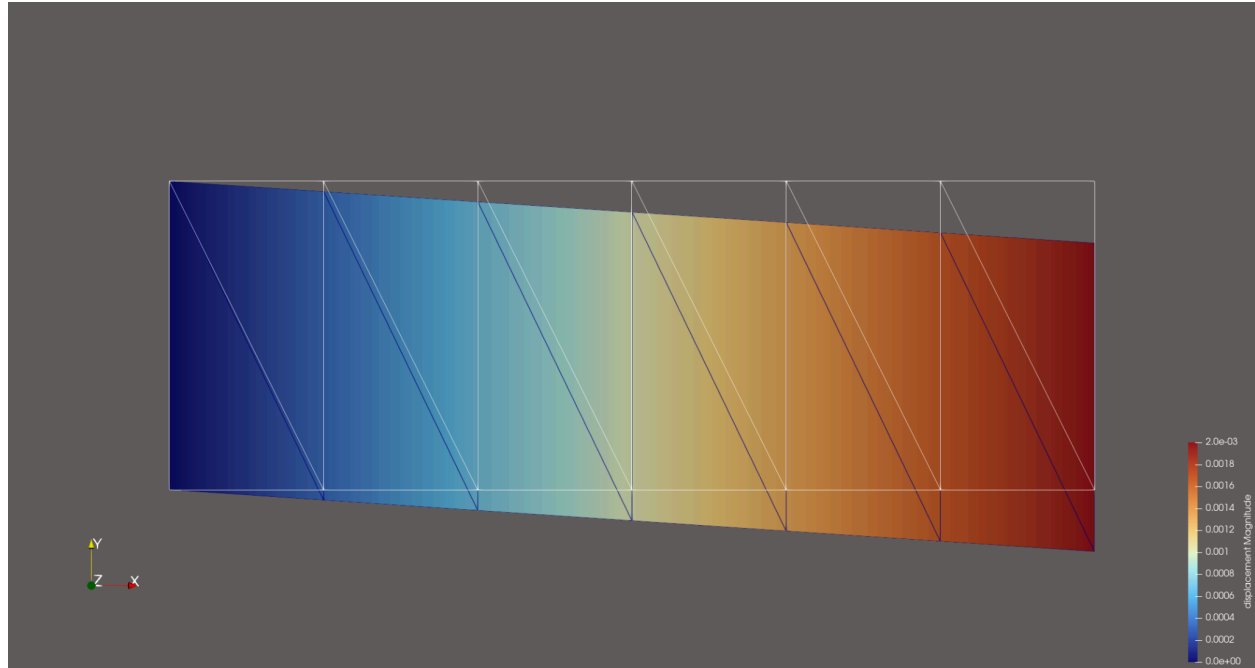


For BCs given in Fig1(c)

Part 3 (a): 12x1 Linear Triangle Elements

#u dofs = 3 ; quadRule = 1

#p dofs = 1 ; quadRule = 1

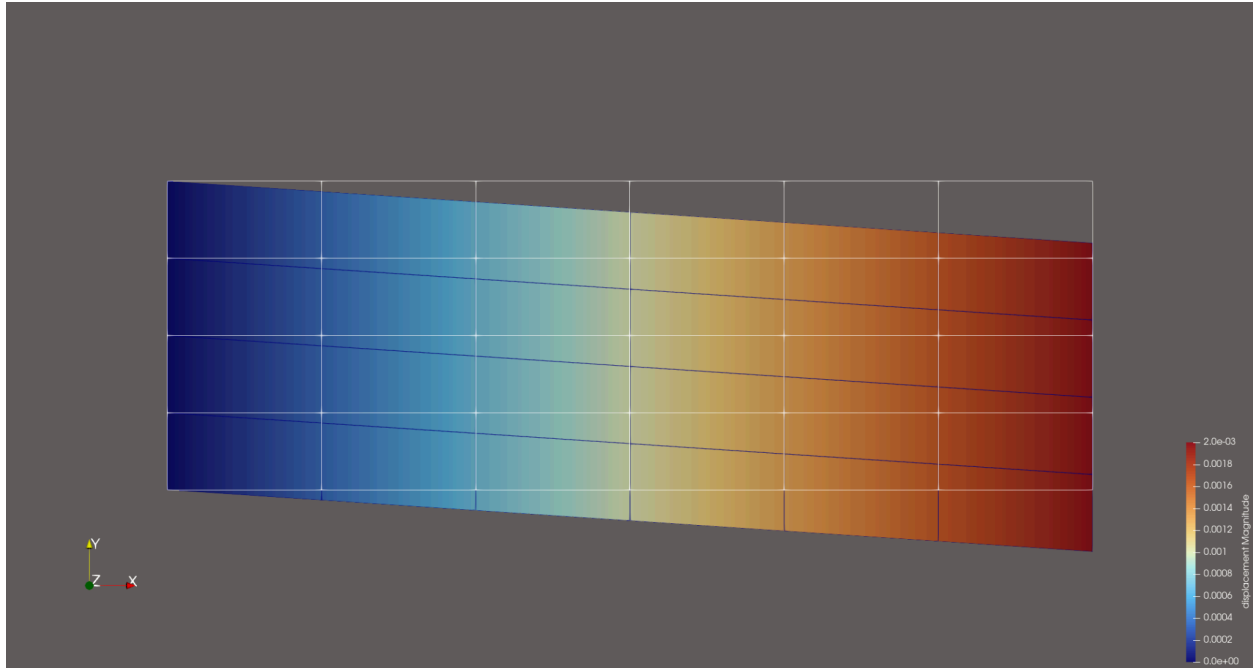


For BCs given in Fig2

Part 3 (b): 6x4 Linear Quadrilateral Elements

#u dofs = 4 ; quadRule = 1

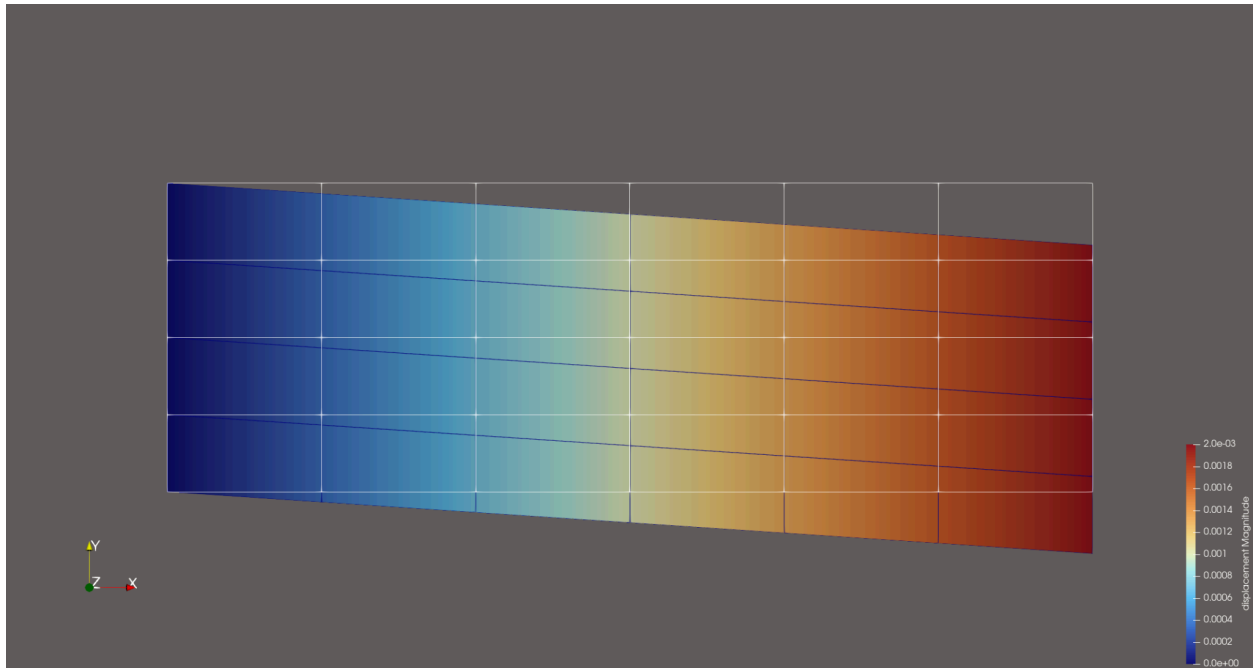
#p dofs = 1 ; quadRule = 1



For BCs given in Fig2

#u dofs = 4 ; quadRule = 2

#p dofs = 4 ; quadRule = 2



For BCs given in Fig2

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#u dofs = 4 ; quadRule = 2

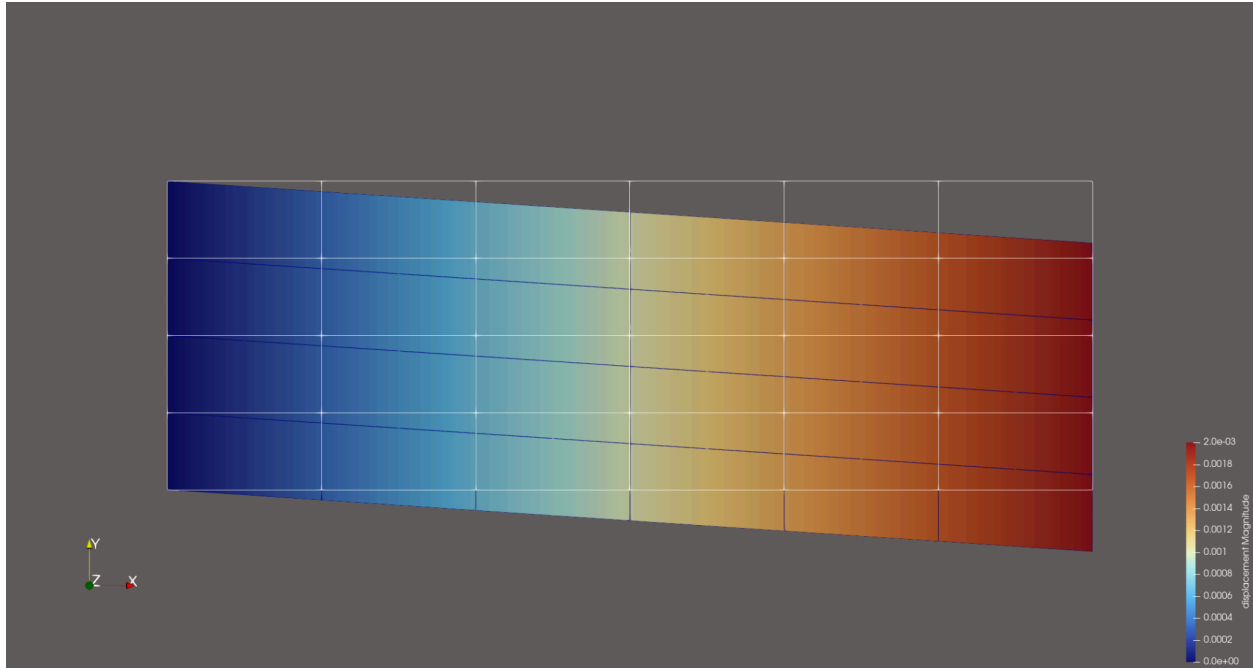
#p dofs = 4 (bubble) ; quadRule = 2

//error - could not resolve in time//

Part 3 (c): 6x4 Quadratic Quadrilateral Elements

#u dofs = 9 ; quadRule = 2

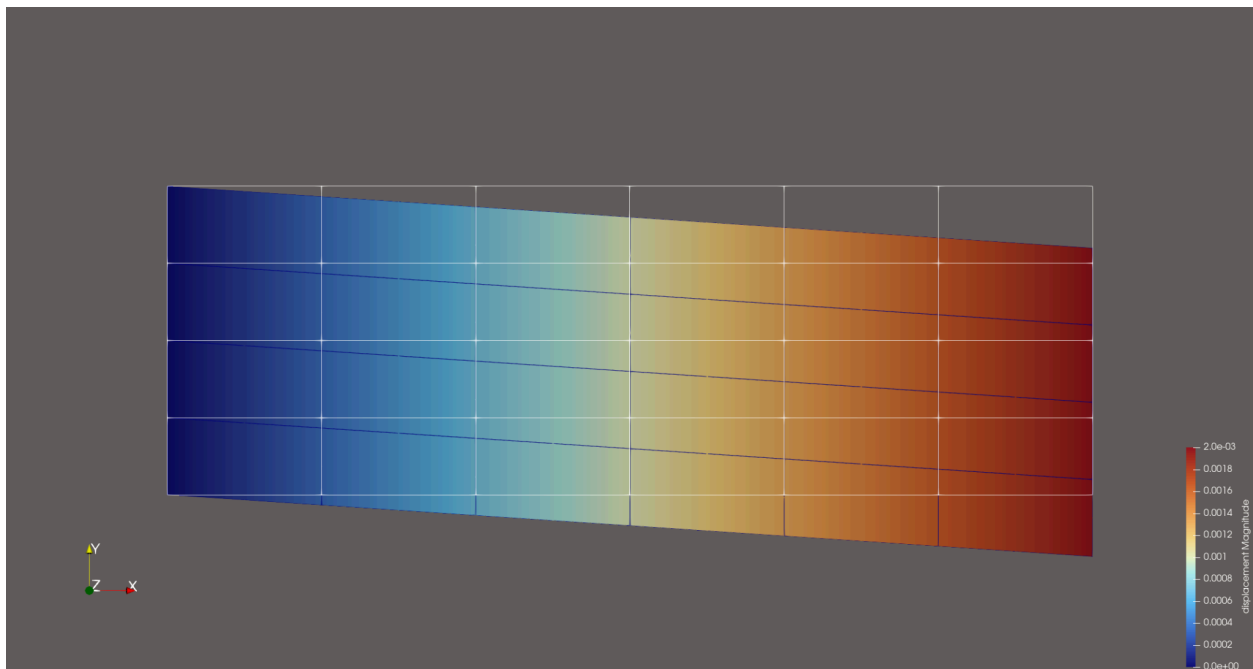
#p dofs = 1 ; quadRule = 1



For BCs given in Fig2

#u dofs = 9 ; quadRule = 2

#p dofs = 4 ; quadRule = 2



For BCs given in Fig2

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```
#u dofs = 9 ; quadRule = 2  
#p dofs = 4 (bubble) ; quadRule = 2  
//error - could not resolve in time//
```